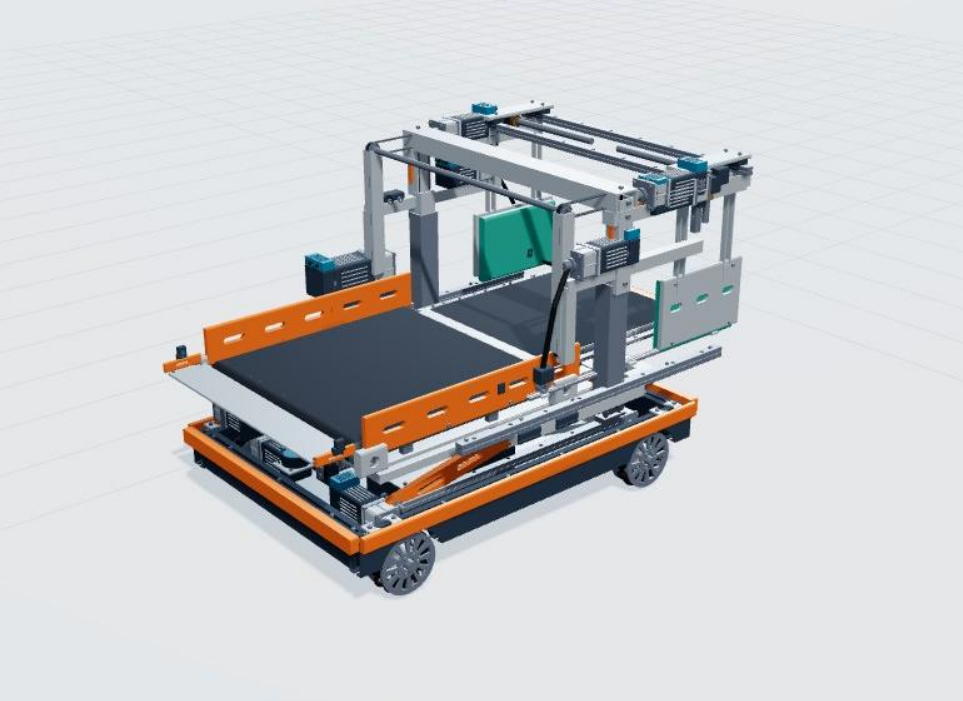
Cargo-Lift Robot

Autonomous Mecanum–Scissor Lift Robot for Loose-Baggage Handling in Narrow-Body Aircraft Cargo Holds



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Design payload	Actuators	Home envelope	Reference endurance
32 kg (event range 20–32 kg)	14 (4 mecanum + 10 handling axes)	1195 x 791 x 805 mm (hold ceiling 1200 mm)	2.26 h on 48 V / 30 Ah pack

1. Concept Overview and System Architecture

A compact mecanum mobile robot receives one bag at a time from an external conveyor at the aircraft side door, drives into the hold and places it directly on the floor or on a lower bag. An open portal keeps all transverse drives, guides and motors above or beside the bag path, so the transfer corridor stays clear. The robot works inside a confined envelope under 1.2 m (home height 805 mm) and never needs a person inside the hold.



Robot working in the generic cargo hold with the external feeder at the side door

- Key innovations
    1. Open-portal head: columns and overhead drives keep the bag path free, giving a 560 mm minimum passage for bags up to 800 x 500 x 320 mm.
    2. Hanging jaws + retractable support blade: two compliant pads grip the bag, a 35 mm micro-lift lets the blade slide under it, so the bag is supported, not only clamped.
    3. Direct stacking and top-down unloading: the robot places or removes bags on the stack itself, with no second handover.
    4. Mecanum base: longitudinal travel, lateral docking and whole-base rotation inside a reserved turning bay, so the aisle never blocks.
    5. Interlocked sequence: lift height is kept until the head has retracted clear of the bag, and the head is retracted before any loaded turn.

Subsystems and their function:

Subsystem	Function in the operating cycle
Mecanum mobile base	Longitudinal travel, lateral docking and whole-base rotation within the reserved turning bay.
Twin scissor lift	Raises the conveyor and head assembly to the required stacking level.
Open portal transfer head	Outboard columns, overhead drives, telescoping Z stages, two jaws and a retractable support blade.
Main and nose conveyors	Receive or return a bag through the open corridor at the side door.
External ground equipment	Feeder terminates at the door; a level service platform is used only for robot entry/exit and parks away during transfer.

- Control and power architecture

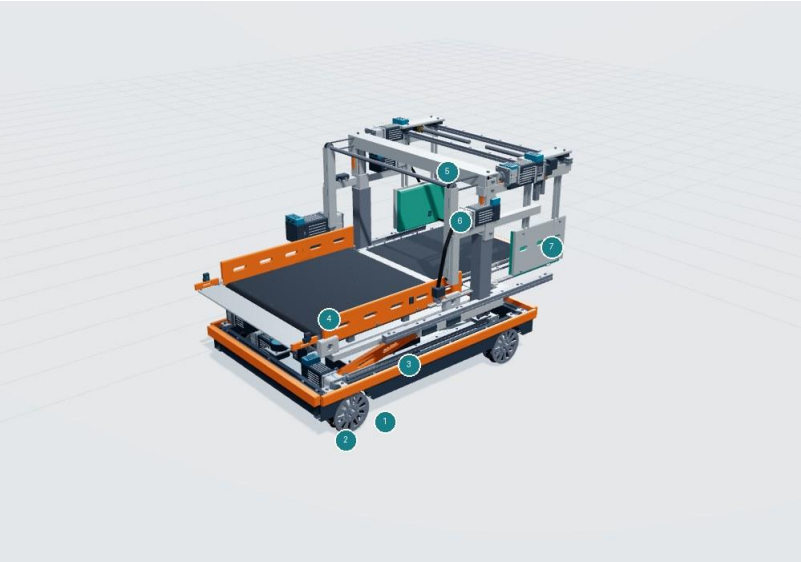
Perception/supervision computer (industrial PC or Jetson-class) → sensors (RGB-D, 2D LiDAR, IMU, photoeyes, jaw load, encoders, limit switches) → motion controller with isolated CANopen (homing, synchronous commands, force caps, watchdog) → 48 V power branch (battery, fuse, pre-charge, contactor, 24/12/5 V DC/DC) → integrated servo drives with brakes on 14 physical motor axes (A01–A14).

2. Mechanism and Key Design Details

Parameter	Design value
Home envelope (L x W x H)	1195 x 791 x 805 mm
Extended pose envelope	1635 x 791 x 1071.5 mm (a defined pose, not the full swept envelope)
Chassis / wheelbase / track	980 x 660 mm chassis; 720 mm wheelbase; 580 mm track
Mass empty / loaded	196 kg / 228 kg at the 32 kg design payload
Mecanum wheels	150 mm diameter, 12 rollers, 45 degree roller angle
Speed / acceleration	Empty 0.30 m/s; loaded 0.15 m/s; limit 0.25 m/s2
Scissor lift	650 mm bars; height 120–430 mm; vertical travel 310 mm
Transfer belt height	366 mm home; 676 mm with the lift fully raised
Head extension / Z stages	440 mm horizontal; 345 mm Z in two 172.5 mm stages
Support blade / micro-lift	560 mm blade retraction; 35 mm jaw micro-lift
Jaws	600 mm pad-to-pad aperture; 220 mm travel per side; pads 250 x 175 mm
Rated bag / passage	Rigid aligned bag up to 800 x 500 x 320 mm; minimum passage width 560 mm
Demonstration bag	650 x 360 x 260 mm; three-layer stacks
Vertical clearance	Lowest overhead member 366.5 mm above belt: 46.5 mm head clearance over a 320 mm bag. Maximum-height configuration 1150 mm, 50 mm under the 1200 mm ceiling.

The passage check includes ±10 mm lateral pose allowance and 10 mm top allowance. It models a rigid aligned bag; straps, soft deformation and tilted oversize items need separate handling rules.

Structure and assemblies:



CAD keys 1–7 mark the separate motion groups:

CAD key	Assembly and construction intent
1 – Lower chassis	Low battery compartments, bolted beams, belly plate, bumpers, accessible drive modules.
2 – Wheel modules	Four independent drives, separate hub bearings, detailed passive roller geometry.
3 – Scissor module	Paired arms, pins, sliding rail blocks, ballscrew-driven slider; independent anti-drop device required.
4 – Conveyor deck	Main belt, end rollers, receiving bridge, guide rails, nose transfer and support blade.
5 – Open portal	Outboard columns and top cross-members keep guides, screws and motors out of the bag path.
6 – Hanging jaws	Two compliant pads, overhead carriages, short micro-lift for blade insertion.
7 – Outboard actuators	Motor/gear housings, timing pulleys, racks, screws, protective brackets.

CAD source: 1771 named components, 2181 solid bodies; motion model 71 links, 70 joints.

3. Actuation and Preliminary Sizing

The 14-actuator budget totals 3800 W mechanical nameplate power (sum of motor classes, not battery draw). Motors are Leadshine iSV2-CAN integrated CANopen servos (candidates).

Axis	Motor class	Transmission and design requirement
A01–A04 mecanum drives	200 W, 48 V	20:1 planetary + 1:1 belt; ~10.9 N.m continuous per wheel; holding brake
A05 scissor lift	750 W, 48 V	6:1 + 25 x 5 mm ballscrew; slider travel ~151.4 mm; thrust support ≥ 15 kN
A06 head Z	750 W, 48 V	20:1, synchronising shaft, timing belts, 2:1 telescopic cascade; 345 mm; 1219 N lift force
A07 head yaw	100 W, 48 V class	150:1 total, slew bearing; ≥ 39 N.m; loaded yaw limited to ±30 degrees
A08 extension	200 W, 48 V	10:1 rack/pinion (20 mm), twin linear rails; 440 mm; force cap 200 N
A09 main belt	200 W, 48 V	20:1, 50 mm roller; 0.20 m/s belt speed; transfer force cap 180 N
A10 nose belt	200 W, 48 V	20:1 plus 2:1 speed increase; 16 mm roller; force cap 180 N
A11 jaw micro-lift	200 W, 48 V	2:1 + bevel + 16 x 5 mm ballscrew; 35 mm; ~1230 N; brake and catcher
A12–A13 jaws (2)	200 W each	16 x 5 mm screws; 220 mm stroke; force limited to 450 N per jaw with load feedback
A14 support blade	200 W, 48 V	10:1 rack/pinion (12 mm); 560 mm stroke; 300 N capacity target; 60 N initial loaded pull limit

Open item: the 6020V48G catalogue page lists 200 W / 0.64 N.m in its feature list but 400 W / 1.27 N.m in its table; the design keeps the conservative 200 W class. Motor brakes do not replace an independent mechanical anti-drop device.

Analytical sizing results:

Mechanism	Assumptions and result
Wheel drive	228 kg, rolling resistance 0.04, 2 degree slope, 0.25 m/s2, efficiency 0.65, factor 1.5: required 9.72 N.m per wheel vs 10.94 N.m available.
Main scissor	115.5 kg supported, minimum angle 10.64 degrees: design screw force 10.41 kN vs 13.79 kN available.
Head Z cascade	74.2 kg lifted: design force 1092 N vs 1219 N theoretical; telescopic overlap 77.5 mm at full travel.
Jaw friction grip	450 N per jaw, friction 0.60: 540 N capacity vs 314 N bag weight (32 kg), static ratio 1.72; pad pressure ~10.3 kPa.
Main conveyor	180 N cap, 25 mm roller: 5.29 N.m required vs 11.52 N.m available.
Extended stability	Head forward, full extension, lift raised, stationary base, 60 N pull: edge moment ratio ~1.30 (limited static case).

- Limits carried into the controller
    1. Base stays stopped during extension, placement and upper-stack pickup; head retracts before whole-base turning.
    2. The loaded base rotates in the reserved bay; the 90 degree loaded head-yaw configuration is excluded because the static edge ratio falls below 1.
    3. Calculations use estimated masses and assumed efficiencies; fatigue life, contact friction, braking distance and floor-panel compatibility are not established.

4. Loading and Unloading Operations

Loading sequence (door to stack):

Step	Action and interlock
1 – Enter	Head stored, lift low, service platform level with hold floor; enter through the open side door; park the platform away before the feeder approaches.
2 – Receive	Dock facing the door; base stationary, jaws open, extension and head Z at home, receiving path clear. Belts run only after transfer-ready verification.
3 – Secure and shuttle	Stop belts, grip with load feedback, confirm support. Retreat into the turning bay, rotate the base 90 degrees, travel toward the target lane.
4 – Place	Stop the base. Lift to stack level, extend, retract the blade, release the bag onto the support below.
5 – Clear and return	Keep lift height while retracting the head; lower only after the portal has cleared the bag; return to the door dock.

- Unloading sequence (stack to door)
    1. Approach the top accessible bag with the base stopped and align the open jaws.
    2. Apply the force-limited grip, raise the micro-lift by up to 35 mm and insert the support blade.
    3. Retract from the stack before lowering the lift; return to the door and stop.
    4. Open the transfer corridor and reverse the belts to the external feeder.
    5. Work top-down, in reverse placement order. The demonstration unloads four bags from the 30-bag filled state.

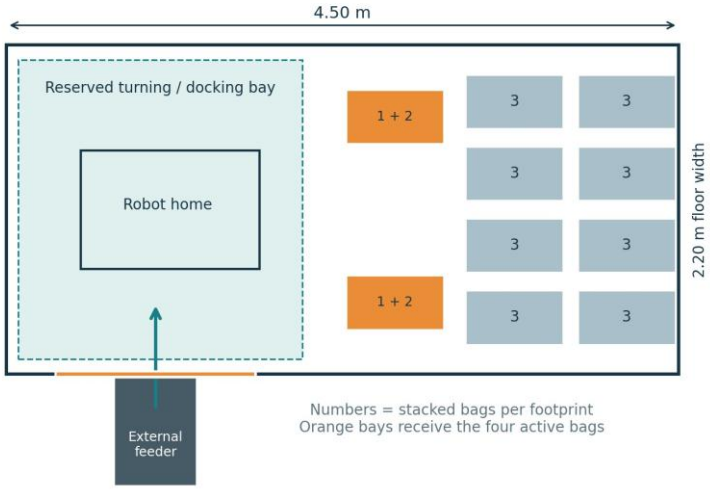
Cycle time and throughput:

Cycle phase	Time [s]	Bus power [W]
leave dock / loaded turn	3 / 4	340 / 650
Loaded travel / lift and stack	5 / 8	600 / 1000
Empty return / empty turn	4 / 2	480 / 500
Dock / wait and verify	2 / 2	270 / 110
Total reference cycle	30 s per bag	avg 610 W

Throughput check against the event target. One robot completes about 2 bags/min (60 / 30 s). The event target of ≥ 12 bags/min is about 6 times higher, which would need roughly 6 robots or a major cycle redesign. This concept does not meet that target as a single unit; it is a baggage-handling and compatibility concept, and this gap is stated openly in Section 7.

Endurance: 1440 Wh x 0.80 x 0.94 x 0.85 = 920.45 Wh usable; 920.45 Wh / 610 W = 1.51 h (1.21 h with 20 % capacity ageing), about 180 bags per charge. Charging: 10 A, about 2.9 h for an 80 % refill. Estimate only, not a discharge test.

5. Narrow-Body Cargo-Hold Compatibility and Space Utilisation



Plan view of the generic hold: reserved turning/docking bay, working aisle and storage footprints (numbers = bags per footprint)

Event requirement	How the concept responds
Working height < 1.2 m	Home height 805 mm; maximum configuration 1150 mm, 50 mm below the 1200 mm ceiling; 46.5 mm head clearance over a 320 mm bag.
Bag mass 20–32 kg	Designed for 32 kg; grip ratio 1.72 at 450 N per jaw (540 N vs 314 N); sizing covers the full 20–32 kg range.
Irregular geometry, material, fragility	Rated for rigid aligned bags up to 800 x 500 x 320 mm; compliant replaceable pads at ~10.3 kPa; force-limited grip with load feedback. Soft bags, straps and tilted oversize items need separate rules and trials.
Floor load ≤ 600 kg/m2	228 kg loaded over the 980 x 660 mm chassis (0.65 m2) averages ~350 kg/m2 (our estimate from report data); about 57 kg per wheel on 150 mm wheels. Concentrated wheel loads must be confirmed on the real floor panels.
Door access	Side-door clear opening 1300 x 1050 mm vs robot width 791 mm; level service platform for entry and exit only.

Hold model and space utilisation:

Parameter	Scene value
Hold interior	4500 mm long, 2200 mm wide, 1200 mm high (generic bulk compartment)
Detailed geometry	Floor panels, tie-down rails, curved liners, formers, bulkheads, door frame and hinges
Bags	26 initial / 30 maximum / 26 final; active bags 650 x 360 x 260 mm
Storage pattern	Eight footprints of three bags at x = 3.40 / 4.15 m; two front footprints receive the four active bags
Aisle and turning bay	Reserved and kept clear throughout; whole-base rotation happens inside the bay

The model is a generic bulk compartment, not an Airbus/Boeing fuselage. Door, liner contours, floor loads, sill geometry and access rules need aircraft-specific data; a clear geometric fit alone does not establish compatibility. 'Filled' means the designated 30-bag storage bays, not the whole volume.

6. Autonomy, Safety and Human Interaction

Sensing and autonomous functions (proposed hardware):

Device	Purpose
2D LiDAR + IMU + wheel encoders	Estimate base motion and check reserved-space clearance; to be validated near reflective liners and feature-poor surfaces.
RGB-D / depth camera	Estimate bag dimensions, grasp region and top surface; detect an unsuitable placement.
Entry/exit photoeyes + height gate	Confirm a bag is inside the corridor and within the planned envelope before belt motion.
Jaw load sensing + motor feedback	Limit clamp force, detect loss of grip; belt current and position monitoring detects a jam.
Dual end limits / position feedback	Homing, hard stops and plausibility checks on lift, extension, blade and telescopic stages.
Supervisor computer	Bag pose, job queue, path planning and operator interface; verifies transfer-ready logic before each phase.

Fault responses and interlocks:

Condition	Required response
Bag exceeds envelope or straps enter a gap	Stop transfer; retain the bag and request operator recovery.
Grip, encoder or jam check fails	Stop the current phase; no blind lift or extension retries.
Obstacle enters the reserved path	Hold the job until the path is revalidated.
Upper-layer placement complete	Retract at the current height before lowering the lift.
Exit requested	Park the external feeder, return the level platform, leave with the head stored.

- Safety measures

1. Independent stop and retention: hardwired stop chain, power contactor, mechanically retained vertical axes and recovery controls. No certified STO function is assumed for the candidate drives.

2. Anti-drop device on the scissor lift is a prototype requirement, motor brakes do not replace it.
3. Force and speed limits: 450 N jaw cap, 180 N belt cap, 200 N extension cap, 60 N blade pull limit, 0.15 m/s loaded travel.
4. Stability rules: base stopped during extension; head retracted before loaded turns; excluded 90 degree yaw case.
5. Human interaction: no person is needed in the hold; operators load at the feeder and handle recovery requests; the robot holds its job whenever the reserved path is obstructed.
6. Separation of areas: the platform and feeder park away during transfer, so robot and ground equipment never share the corridor.

Interlocks describe the intended physical controller. The animation follows predetermined successful poses and does not demonstrate sensor-driven fault recovery. Connector pinouts, bonding, cable sizing and functional-safety hardware still need engineering.

7. Advantages, Limitations, Validation and Next Steps

Advantages	Limitations
Low 805 mm profile fits the 1.2 m hold; open portal keeps the bag path and aisle free.	Throughput 2 bags/min per robot (30 s cycle) vs the ≥ 12 bags/min target, need 6 robots to reach the target.
Direct stacking and top-down unloading with blade-supported bags.	Stability shown only in a limited static case; 90 degree loaded yaw excluded.
Compact 1195 x 791 mm, 1.51 h endurance, mecanum in-place rotation.	Rated for rigid aligned bags; soft, strapped or oversize bags need rules and trials.
Detailed STEP models, sizing margins and defined interlocks.	Animation is prescribed motion; no physics, contact, SLAM or autonomous packing validation.
Modular servo-class actuators with force-limited, load-sensed handling.	Aircraft-specific door, sill and floor data not yet applied; motor catalogue data inconsistent for one variant.

- Before manufacturing release

1. Confirm one aircraft type and obtain door, sill, hold and floor-load data.
2. Complete tolerances, cable/shaft routing, screw buckling and bearing life checks, structural analysis and prototype tests.
3. Measure real bag friction and grip damage; trial soft bags and loose straps; validate stop and recovery behaviour.
4. Resolve IP20 motor enclosure, contamination and cooling; plan service access for belts, rollers, screws, jaw pads and battery isolation.